

AWS Invoice Pipeline — API Reference

This document provides a comprehensive, code-driven reference of all API Gateway endpoints deployed for the AWS Invoice Processing Pipeline. All REST API endpoints are secured via AWS API Gateway and route to either the `invoice-ingestion` Lambda (for uploads) or the `approval-handler` Lambda (for approvals and data retrieval).

Base Architecture

- **Protocol:** HTTPS / REST
- **CORS:** Enabled for all origins (*), supporting GET , POST , OPTIONS .
- **Base URL (Example):** `https://<api-id>.execute-api.<region>.amazonaws.com/<env>`
- **Content-Type:** `application/json` (except for email callback endpoints which return `text/html`).

1. Ingestion Endpoints

Handled by the `invoice-ingestion` Lambda function.

POST /upload

Directly uploads an invoice document (PDF) into the pipeline.

- **Headers:**
 - **Content-Type :** `multipart/form-data` OR `application/pdf` (base64 encoded via API Gateway)
 - **X-Filename :** (Optional) Original filename of the upload
 - **X-Invoice-Metadata :** (Optional) JSON string containing manual form fields from the UI (Vendor, PO Number, etc.)
- **Request Body:** The raw binary PDF file or a base64 encoded string if sent directly via API Gateway proxy.
- **Success Response (200 OK):**

```
{
  "success": true,
  "invoiceId": "INV-2026-A1B2C3D4",
  "message": "Invoice uploaded and pipeline started"
}
```

- **Error Responses:**
 - **400 Bad Request :** "No file data provided"
 - **405 Method Not Allowed :** "Method not allowed"

2. Email Callback Endpoints

Handled by the `approval-handler` Lambda function. These endpoints are triggered via hyperlinks injected into SES emails.

GET /approve

Approves an invoice directly from an email notification.

- **Query Parameters:**
 - `token` (Required): Step Functions task token.
 - `invoiceId` (Required): ID of the invoice.
- **Success Response (200 OK):** Returns a rendered HTML page confirming the invoice was approved. The Step Functions workflow resumes with a `SendTaskSuccess` command.

GET /reject

Rejects an invoice directly from an email notification.

- **Query Parameters:**
 - `token` (Required): Step Functions task token.
 - `invoiceId` (Required): ID of the invoice.
- **Success Response (200 OK):** Returns a rendered HTML page confirming the invoice was rejected. The Step Functions workflow resumes with a `SendTaskFailure` command.

3. Dashboard & Operations Endpoints

Handled by the `approval-handler` Lambda function. These are consumed by the React (Vite) frontend.

GET /dashboard

Retrieves aggregated statistics across the entire database to power dashboard KPIs.

- **Query Parameters:** None.
- **Success Response (200 OK):**

```
{
  "totalInvoices": 150,
  "processed": 120,
  "inProgress": 10,
  "exceptions": 15,
  "inReview": 3,
  "pendingReview": 2,
  "resolved": 0,
  "processedPercentage": "80.0",
  "exceptionPercentage": "10.0"
}
```

GET /invoices

Retrieves a paginated list of all invoices.

- **Query Parameters:**

- `status` (Optional): Filter by `InvoiceStatus` (e.g., `PROCESSED` , `IN_REVIEW`).
- `pageSize` (Optional, Default `25`): Limit results per page.
- `lastKey` (Optional): JSON string representation of DynamoDB `LastEvaluatedKey` for pagination.

- **Success Response (200 OK):**

```
{
  "items": [
    {
      "invoiceId": "INV-...",
      "vendorName": "Acme Corp",
      "totalAmount": 1000.00,
      "status": "PROCESSED"
    }
  ],
  "total": 1,
  "hasMore": false
}
```

GET /invoices/{invoiceId}

Retrieves complete details of a specific invoice along with its audit history.

- **Path Parameters:**

- `invoiceId` (Required): The ID of the target invoice.

- **Success Response (200 OK):**

```
{
  "invoice": {
    "invoiceId": "INV-...",
    "vendorName": "Acme Corp",
    "lineItems": [],
    "anomalies": []
  },
  "auditLogs": [
    {
      "eventType": "INGESTION",
      "event": "Invoice received via email",
      "timestamp": "2026-06-25T12:00:00Z"
    }
  ]
}
```

GET /exceptions

Retrieves all invoices that have encountered anomalies or require manual review.

- **Query Parameters:**

- `pageSize` (Optional, Default `25`): Pagination limit.
- **Success Response (200 OK):** Returns a combined list of invoices with statuses `EXCEPTION`, `IN_REVIEW`, `PENDING_REVIEW`, and `RESOLVED`.

```
{
  "items": [],
  "total": 0,
  "stats": {
    "totalExceptions": 0,
    "pendingReview": 0,
    "inProgress": 0,
    "resolved": 0
  }
}
```

GET /audit

Queries the audit log table directly for a specific invoice's history.

- **Query Parameters:**
 - `invoiceId` (Required): The target invoice ID.
- **Success Response (200 OK):**

```
{
  "entries": [
    {
      "auditId": "...",
      "eventType": "VALIDATION",
      "details": "..."
    }
  ]
}
```

4. Workflow Actions

POST /approvals

Submits a review decision (Approve, Reject, or Reprocess) from the dashboard UI. Handles dynamic updates to extracted fields (like fixing an amount or GSTIN manually).

- **Request Body:**

```
{
  "invoiceId": "INV-2026-...",
  "action": "APPROVE",
  "taskToken": "AAAAKgAAAAIA...",
  "reviewer": "Admin User",
  "comments": "Fixed subtotal mismatch",
}
```

```
"correctedFields": {  
  "INV-2026-...-subtotal": 500.00  
}  
}
```

- **Behavior:**

- **APPROVE** : Updates DB fields using `correctedFields` . Sets status to `PROCESSED` . Resumes Step Functions workflow via `SendTaskSuccess` . Clears anomalies.
- **REJECT** : Sets status to `RESOLVED` . Logs rejection comments. Resumes workflow via `SendTaskFailure` .
- **REPROCESS** : Reverts status to `IN_PROGRESS` . (Allows manual trigger back to extraction).

- **Success Response (200 OK):**

```
{  
  "success": true,  
  "invoiceId": "INV-2026-...",  
  "action": "APPROVE",  
  "message": "Invoice approved successfully"  
}
```